Dec-ZoomSIB-Relay without a server: proofs

Lower bound, regret, communication, robustness and agreement

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Abstract

We analyse the serverless version of Fed-ZoomSIB, where N agents communicate only along the edges of a graph $\mathcal{G}$. Every message is flooded, so it reaches every agent within Λ rounds; on a static connected graph Λ is the diameter D. We prove five results. (1) No serverless algorithm beats $\Omega((NT)^{2/3})$ (Theorem 5). (2) Dec-ZoomSIB-Relay has regret equal to the server bound plus an additive $O(L_f\, N\Lambda\, N_{\mathrm{b}})$, that is, $\tilde{O}\big((dNT)^{2/3} + N\Lambda(NT)^{1/3}\big)$ (Theorem 16). (3) It sends $O\big(|E|\, N\, N_{\mathrm{b}}^2 \log_{1+\gamma}(NT)\big)$ scalars, at most logarithmic in T (Theorem 21). (4) Time-varying, unreliable and directed graphs only change Λ (Corollary 24). (5) If agents freeze different directions, the regret grows by at most $4L_{f'}X_{\max}\,\eta\,NT$, where η is the worst direction error, and the cost is zero when Phase 1 pools exactly (Theorem 26). The analysis uses *synchronised epochs*: all agents act on the same delayed pooled table plus their own recent pulls. This keeps the confidence argument of the server case unchanged and removes the $\sqrt{N}$ loss of the earlier sketch.

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1 Setting, notation and imported results

Bandit model. There are N agents and T rounds. At round t agent i observes K contexts $X_{i,t} = (x_{i,t,1}, \dots, x_{i,t,K})$, drawn i.i.d. from $\mathcal{D}$ on $\mathbb{R}^d$, independently across agents and rounds. It pulls an arm $a_{i,t}$ and observes

$$y_{i,t} = f(\langle x_{i,t,a_{i,t}}, \theta^* \rangle) + \eta_{i,t}.$$

Both θ^* (with $\|\theta^*\|_1 = 1$) and f are unknown and shared by all agents. Write $z = \langle x, \theta^* \rangle$ for the *true index* of an arm x. The network regret is

$$R_T(N) = \sum_{i=1}^{N} \sum_{t=1}^{T} r_{i,t}, \qquad r_{i,t} = \max_{a \le K} f(\langle x_{i,t,a}, \theta^* \rangle) - f(\langle x_{i,t,a_{i,t}}, \theta^* \rangle).$$

Order of pulls. Within a round, all N agents act in parallel. For the analysis we order all pulls *lexicographically* by (round, agent): $(t, i) \prec (t', i')$ iff $t < t'$, or $t = t'$ and $i < i'$. Let $\mathcal{F}_{t,i}$ be the σ-field generated by all contexts, actions and rewards strictly before (t, i) in this order, together with $X_{i,t}$ and the algorithm's internal randomness.

Assumption 1 (DBG Assumptions 1–3)**.**
(i) $\eta_{i,t}$ is independent of the contexts and conditionally σ-sub-Gaussian: $\mathbb{E}[e^{\lambda \eta_{i,t}} \mid \mathcal{F}_{t,i}, a_{i,t}] \le e^{\lambda^2 \sigma^2 / 2}$ for all $\lambda \in \mathbb{R}$.
(ii) $\mathbb{E}[S_j(X)^2] \le M$ for the score $S = -\nabla \log p$.
(iii) Each coordinate of X is κ-sub-Gaussian. On $|z| \le X_{\max}$ we have $|f| \le L_f$ and $|f'| \le L_{f'}$, where $X_{\max} = \kappa \sqrt{2 \log(2dNTK/\delta)}$. With probability $\ge 1 - \delta$, every one of the NTK contexts satisfies $\|x\|_\infty \le X_{\max}$. Call this event E_x.
(iv) $\mu^* = \mathbb{E}[f'(\langle X, \theta^* \rangle)] > 0$.

Communication model. The agents are the vertices of a graph $\mathcal{G} = (V, E)$ with $|V| = N$. Each round has two steps: a *decision step*, in which every agent pulls an arm, and then a *communication step*, in which every agent may send messages to its neighbours. A message sent in the communication step of round s is received before the decision step of round $s + 1$. *Flooding* means that an agent forwards every message the first time it receives it, to all its neighbours, and never again.

Assumption 2 (Delivery time)**.** There is a known integer $\Lambda \ge 1$ such that a message that is flooded from any agent in the communication step of round s has reached every agent by the end of the communication step of round $s + \Lambda - 1$. Each agent knows N, Λ and the round number.

Lemma 1 (Flooding on a static graph)**.** *If $\mathcal{G}$ is static, connected and has diameter D, Assumption 2 holds with $\Lambda = D$.*

Proof. Let agent a flood a message in step s. We show by induction on $h \ge 1$ that every agent at distance h from a holds the message after step $s + h - 1$. For $h = 1$: neighbours of a receive it in step s. For the inductive step, let v be at distance $h + 1$. It has a neighbour u at distance h, which holds the message after step $s + h - 1$. If u received it in step $s + h - 1$ or earlier, it forwards it no later than step $s + h$. So v holds it after step $s + h$. Every agent is at distance at most D. □

Imported results. The following results are proved elsewhere and used as black boxes. Here DBG refers to Dey, Bhore & Ghosh (2026).

Lemma 2 (Stein estimator; DBG Lemma B.1 with Lemma 2.1)**.** *Let $\hat{\theta} = \frac{1}{n}\sum_{s=1}^{n} \varphi_\tau(y_s S(x_s))$ be the truncated Stein estimator on n i.i.d. pairs, with $\tau = \sqrt{3(\sigma^2 + L_f^2)Mn/\log(2d/\delta)}$, and let $\hat{\theta}_0 = \hat{\theta}/\|\hat{\theta}\|_1$. If $C_\theta d\sqrt{\log(2d/\delta)/n} \le \mu^*/2$, then with probability $\ge 1-\delta$,*

$$\|\hat{\theta}_0 - \theta^*\|_1 \le \varepsilon_0(n) := \frac{4C_\theta d}{\mu^*}\sqrt{\frac{\log(2d/\delta)}{n}}, \qquad C_\theta = \tfrac{4\sqrt{3}}{3} + 2\sqrt{2}\sqrt{M(\sigma^2 + L_f^2)}.$$

Lemma 3 (Single-agent lower bound; DBG Theorem 5.1)**.** *There are $c > 0$ and n_0 such that for every horizon $n \ge n_0$ and every single-agent policy π, $\sup \mathbb{E}[R_n(\pi)] \ge c\, n^{2/3}$. The supremum is over instances that satisfy the regularity assumptions (with $d = 1$, Gaussian contexts, and a number of arms that may grow logarithmically with n).*

Lemma 4 (Martingale concentration; Azuma–Hoeffding for sub-Gaussian increments)**.** *Let $(\xi_m)_{m\ge 1}$ be adapted to a filtration $(\mathcal{H}_m)$ and satisfy $\mathbb{E}[e^{\lambda \xi_m} \mid \mathcal{H}_{m-1}] \le e^{\lambda^2\sigma^2/2}$ for every $\lambda \in \mathbb{R}$. Then for every fixed $p \ge 1$ and $\delta' \in (0,1)$,*

$$\mathbb{P}\Big(\Big|\sum_{m=1}^{p} \xi_m\Big| > \sigma\sqrt{2p\log(2/\delta')}\Big) \le \delta'.$$

2 Lower bound

Theorem 5 (No serverless algorithm beats $(NT)^{2/3}$)**.** *Let $\mathcal{A}$ be any N-agent algorithm, on any graph and with any communication. This includes the extreme case where every agent sees every other agent's data after every round. Then for $NT \ge n_0$,*

$$\sup_{instances} \mathbb{E}\big[R_T(N;\mathcal{A})\big] \;\ge\; c\,(NT)^{2/3}.$$

Proof. The proof is a reduction: from $\mathcal{A}$ we build a single-agent policy π with horizon $n = NT$ whose regret has the same law as the network regret of $\mathcal{A}$. Then we apply Lemma 3.

Step 1: relabelling time. Map the single-agent round $u \in \{1,\dots,NT\}$ to the pair

$$u \;\longleftrightarrow\; (t,i), \qquad u = (t-1)N + i, \quad t \in [T],\ i \in [N].$$

Step 2: the policy π. Policy π stores the complete state of $\mathcal{A}$: every agent, every message buffer, and every random seed. It draws the seeds itself. At its round $u = (t,i)$:

(a) it receives the context set X_u and gives it to the simulated agent i;
(b) the simulated agent i chooses a_u from its state after round $t-1$, using exactly the rule of $\mathcal{A}$;
(c) π plays a_u and stores the reward y_u without giving it to the simulation;
(d) if $i = N$, π gives the N stored rewards of round t to the simulated agents and runs one communication step of $\mathcal{A}$.

Each action a_u depends only on X_u, on rewards of rounds $u' \le (t-1)N < u$, and on π's own randomness. So π is a valid single-agent policy.

Step 3: the laws coincide. Let H_t be the history (contexts, actions, rewards) of rounds $1,\dots,t$ of $\mathcal{A}$, and H'_t the history of rounds $1,\dots,tN$ of π, grouped in blocks of N. We show by induction on t that $H'_t \overset{d}{=} H_t$. The case $t = 0$ is trivial. Assume $H'_{t-1} \overset{d}{=} H_{t-1}$.

- In both systems the N context sets of round t are i.i.d. from $\mathcal{D}^K$ and independent of the past.
- Given the past and the contexts, the N actions are the same measurable function of these and of the seeds, in both systems. This is because π withholds the round-t rewards.

- Given the actions, each reward is $f(\langle x, \theta^* \rangle) + \eta$, with noise of the same conditional law in both systems.

So $H'_t \overset{d}{=} H_t$. The regret is the same function of the history in both systems, namely the sum of the NT per-pull gaps. Hence $R_{NT}(\pi) \overset{d}{=} R_T(N; \mathcal{A})$ on every instance, and

$$\sup \mathbb{E} R_T(N; \mathcal{A}) = \sup \mathbb{E} R_{NT}(\pi) \;\geq\; c\,(NT)^{2/3}$$

by Lemma 3. □

Remark 6. The proof uses only two facts: the agents are statistically identical (they share θ^*, f and $\mathcal{D}$), and within a round they act in parallel. It does not depend on $\mathcal{G}$. A lower bound that grows with the graph (an additive $\Omega(\cdot\, \Lambda)$ term) would need a different argument; this is left open.

3 The algorithm as analysed

The algorithm has a grid of bins on the index line and two phases. The table below fixes the notation.

Symbol	Meaning
T_0	exploration rounds per agent; Phase 2 starts at round $T_1 + 1$, $T_1 = T_0 + \Lambda$
$W = X_{\max}$, Δ	window and bin width; bins $I_j = (-W + (j-1)\Delta,\ -W + j\Delta]$, $j \leq N_{\mathrm{b}} = \lceil 2W/\Delta \rceil$
$L = 2\Lambda$	slot length; slot boundaries $\beta_m = T_1 + mL$, $m \geq 0$
$\mathcal{E}$	set of epochs (a subset of slot boundaries); $\beta_0 \in \mathcal{E}$
$O_{k,j}(s)$	number of Phase-2 pulls by agent k into bin j in rounds $\leq s$
$G_j(s) = \sum_k O_{k,j}(s)$	network count of bin j up to round s (similarly the reward sum $G_j^S(s)$)
$e(t)$	reference epoch at round t, defined in (2)

Phase 1 (rounds $1, \ldots, T_1$). Every agent pulls uniformly at random in rounds $1, \ldots, T_1$. After round T_0, agent k floods the record

$$b_k = \sum_{t \leq T_0} \varphi_\tau\big(y_{k,t} S(x_{k,t,a_{k,t}})\big) \in \mathbb{R}^d,$$

with τ set for the pooled size $n = NT_0$. By Assumption 2, every agent holds all N records before round $T_1 + 1$. Every agent then computes the same direction

$$\hat{\theta}_0 = \frac{\sum_k b_k}{\| \sum_k b_k \|_1}.$$

Pulls in rounds $T_0 < t \leq T_1$ are uniform and are not used.

Bins. An arm x belongs to bin $b(x)$, the bin with $\langle x, \hat{\theta}_0 \rangle \in I_{b(x)}$.

Phase 2: synchronised epochs. Every agent keeps a log of its own Phase-2 pulls, so it can compute $O_{k,\cdot}(s)$ and $O^S_{k,\cdot}(s)$ for any past round s.

(a) *Trigger.* In the communication step of a slot boundary β_m ($m \geq 1$), agent l raises an *alarm* if, for some bin j,

$$\delta_{l,j}(\beta_m) := O_{l,j}(\beta_m) - O_{l,j}(\epsilon_m) \;>\; \gamma \max\big(G_j(\epsilon'_m), 1\big), \tag{1}$$

where $\epsilon_m = \max\{e \in \mathcal{E} : e \leq \beta_{m-1}\}$ and $\epsilon'_m = \max\{e \in \mathcal{E} : e \leq \beta_{m-2}\}$ (with $\beta_{-1} := \beta_0$). In words: l's own pulls into bin j since the last epoch exceed γ times the network count it is currently acting on. The alarm (one scalar: m) is flooded. If at least one agent raises an alarm at β_m, then $\beta_m \in \mathcal{E}$.

(b) *Snapshot.* When agent k first learns that $\beta_m \in \mathcal{E}$, it floods its *snapshot*: the bins j with $O_{k,j}(\beta_m) \neq O_{k,j}(\epsilon_m)$, each with the pair of increments, plus the epoch label.

(c) *Acting.* At round t in slot $m+1$ (that is, $\beta_m < t \leq \beta_{m+1}$), agent i uses the reference epoch

$$e(t) = \max\{e \in \mathcal{E} :\; e \leq \beta_{m-1}\}, \tag{2}$$

which is the same for all agents. It acts on the table

$$n_{i,j}(t) = G_j(e(t)) + \big[O_{i,j}(t-1) - O_{i,j}(e(t))\big], \qquad S_{i,j}(t) = G^S_j(e(t)) + \big[O^S_{i,j}(t-1) - O^S_{i,j}(e(t))\big].$$

Among the bins its arms fall into, it picks $J = \arg\max_j \mathrm{UCB}_{i,j}(t)$, where $\mathrm{UCB}_{i,j} = S_{i,j}/n_{i,j} + w(n_{i,j})$ and $w(0) = \infty$, and plays any arm in bin J.

The confidence width is

$$w(n) = c_\delta \sqrt{2/n}, \qquad c_\delta = \sigma\sqrt{2\log\big(4NN_{\mathrm{b}}(NT)^2/\delta\big)}.$$

Lemma 7 (The protocol is well defined)**.** *(i) In the communication step of β_m, every agent knows ϵ_m, ϵ'_m and $G(\epsilon'_m)$.*

(ii) Before round $\beta_m + 1$, every agent knows $G(e)$ and $G^S(e)$ for each epoch $e \leq \beta_{m-1}$.

Proof. An alarm for $\beta_{m'}$ is flooded in step $\beta_{m'}$ and reaches everyone by step $\beta_{m'} + \Lambda - 1$. Each snapshot is flooded at the latest in step $\beta_{m'} + \Lambda$ and reaches everyone by step $\beta_{m'} + 2\Lambda - 1$. Since $2\Lambda = L$, both are complete before round $\beta_{m'} + L + 1 = \beta_{m'+1} + 1$.

For (ii), take $m' \leq m-1$: everything about epoch $\beta_{m'}$ is known before round $\beta_{m'+1}+1 \leq \beta_m+1$.

For (i), whether $\beta_{m'} \in \mathcal{E}$ is known by step $\beta_{m'} + \Lambda - 1 < \beta_m$ for all $m' \leq m-1$, so ϵ_m and ϵ'_m are known. $G(\epsilon'_m)$ is known before round $\beta_{m-1} + 1$, by (ii) applied to $m-1$. Finally, every agent can sum snapshots to get $G(e)$, because $G_j(e) - G_j(e_{\mathrm{prev}})$ is the sum over k of the increments in the snapshots for e. □

Remark 8 (Relation to the code)**.** The flooding code (`src/dec/phase2/flood.py`) works differently. Each agent republishes its own table when its *own* trigger fires, and each agent acts on the freshest copy of every record it holds. That uses more information, but it changes the confidence argument. In the code, whether a pull of agent k is in agent i's view depends on when k next publishes, which is decided *after* the pull. Agent i's view is then a union of N per-agent prefixes. Bounding each prefix separately costs a factor $\sqrt{N}$ in the width (this was the gap in the earlier sketch). The epoch protocol above avoids this: every agent's view is one common prefix plus its own pulls (Lemma 12). E23 measures the code version. Implementing the epoch version and checking that it behaves the same is a natural next experiment.

4 Regret upper bound

4.1 Good events and bin geometry

Throughout this section, agent k may use its own direction $\hat{\theta}_k$; the protocol above has $\hat{\theta}_k = \hat{\theta}_0$ for every k. This generality is used again in Section 7. Define the *projection error*

$$\varepsilon_{\mathrm{p}} = X_{\max} \max_{k \le N} \|\hat{\theta}_k - \theta^*\|_1.$$

Lemma 9 (Phase 1 is exactly centralised)**.** *$\sum_k b_k/(NT_0)$ is the truncated Stein estimator on $n = NT_0$ i.i.d. samples. Hence, if NT_0 satisfies the sample-size condition of Lemma 2, there is an event E_θ with $\mathbb{P}(E_\theta) \ge 1 - \delta$ on which $\|\hat{\theta}_0 - \theta^*\|_1 \le \varepsilon_0(NT_0)$.*

Proof. Truncation φ_τ acts on each summand separately, with the same τ at every agent, so $\sum_k b_k = \sum_k \sum_{t \le T_0} \varphi_\tau(y_{k,t} S(x_{k,t}))$ is the sum over the pooled sample. Each pull is uniform over the K i.i.d. contexts, so $(x_{k,t}, y_{k,t})$ are i.i.d. across (k,t) with the law of a single-agent Phase-1 sample. Apply Lemma 2 with $n = NT_0$. □

Let $\tilde{I}_j = \big(I_j + [-\varepsilon_{\mathrm{p}}, \varepsilon_{\mathrm{p}}]\big) \cap [-X_{\max}, X_{\max}]$ be bin j widened by ε_{p} on each side, and

$$m_j = \sup_{z \in \tilde{I}_j} f(z), \qquad \omega = L_{f'}(\Delta + 2\varepsilon_{\mathrm{p}}).$$

Lemma 10 (Bin geometry)**.** *On E_x, for every agent k and every arm x it sees:*
(i) x falls in some bin, i.e. $|\langle x, \hat{\theta}_k \rangle| \le W$;
(ii) if x is in bin j under $\hat{\theta}_k$, then its true index $z = \langle x, \theta^* \rangle$ lies in $\tilde{I}_j$;
(iii) hence $m_j - \omega \le f(z) \le m_j$.

Proof. (i) $|\langle x, \hat{\theta}_k \rangle| \le \|x\|_\infty \|\hat{\theta}_k\|_1 = \|x\|_\infty \le X_{\max} = W$. (ii) By Hölder, $|\langle x, \hat{\theta}_k \rangle - z| \le \|x\|_\infty \|\hat{\theta}_k - \theta^*\|_1 \le \varepsilon_{\mathrm{p}}$. Also $|z| \le \|x\|_\infty \|\theta^*\|_1 \le X_{\max}$. (iii) $f(z) \le m_j$ by definition. The interval $\tilde{I}_j$ has length at most $\Delta + 2\varepsilon_{\mathrm{p}}$, and f is $L_{f'}$-Lipschitz on it, so $f(z) \ge m_j - \omega$. □

4.2 Confidence

For each bin j, list all Phase-2 pulls into bin j in lexicographic order, and let $\mu_s = f(z_s)$ and η_s be the mean and the noise of the s-th such pull.

Lemma 11 (What an agent sees)**.** *The pulls counted in $n_{i,j}(t)$ are the union of two sets:*
(i) a common prefix*: all pulls into bin j in rounds $\le e(t)$, i.e. the first $G_j(e(t))$ pulls of bin j's lexicographic list;*
(ii) an own segment*: agent i's own pulls into bin j in rounds $e(t) < t' \le t - 1$, i.e. a contiguous block of agent i's own bin-j pulls.*

Proof. Immediate from the definition of $n_{i,j}(t)$ and $G_j(e) = \sum_k O_{k,j}(e)$. A pull in round $t' \le e(t)$ precedes, in lexicographic order, every pull in a later round. □

Lemma 12 (Confidence)**.** *There is an event E_η with $\mathbb{P}(E_\eta) \ge 1 - \delta$ on which, for every agent i, bin j and Phase-2 round t with $n_{i,j}(t) \ge 1$,*

$$\Big| \frac{S_{i,j}(t)}{n_{i,j}(t)} - \frac{1}{n_{i,j}(t)} \sum_{s \in \mathrm{seen}_{i,j}(t)} \mu_s \Big| \le w\big(n_{i,j}(t)\big),$$

where $\mathrm{seen}_{i,j}(t)$ is the set of pulls counted in $n_{i,j}(t)$.

Proof. By Lemma 11, $S_{i,j} - \sum_{\text{seen}} \mu_s = P + Q$, where P is the noise sum over a prefix of length p of bin j's list, and Q is the noise sum over a contiguous block of length q of agent i's own bin-j pulls, with $p + q = n_{i,j}$.

Prefixes. Let ξ_m be the noise of the m-th pull into bin j, and $\mathcal{H}_m$ the information up to and including the choice of that pull. Whether a given pull goes to bin j is decided from $\mathcal{F}_{t,i}$, so (ξ_m) is a martingale difference sequence with σ-sub-Gaussian increments by Assumption 1(i). If bin j receives fewer than m pulls, pad the sequence with independent $\mathcal{N}(0, \sigma^2)$ variables; this changes no actual sum. Lemma 4 applies for each fixed $p \le NT$.

Own blocks. The same argument applies to the subsequence of agent i's bin-j pulls, and to the sum from its $(a+1)$-th to its b-th element, for each fixed pair $0 \le a < b \le T$.

Union bound. There are at most $N_{\mathrm{b}} \cdot NT$ prefixes and $N \cdot N_{\mathrm{b}} \cdot T^2$ own blocks, in total at most $2NN_{\mathrm{b}}(NT)^2$ events. Applying Lemma 4 to each with $\delta' = \delta/(2NN_{\mathrm{b}}(NT)^2)$, with probability $\ge 1 - \delta$ all of them satisfy

$$|P| \le c_\delta \sqrt{p}, \qquad |Q| \le c_\delta \sqrt{q}.$$

Since $\sqrt{p} + \sqrt{q} \le \sqrt{2(p+q)}$, we get $|P + Q| \le c_\delta \sqrt{2n_{i,j}}$. Dividing by $n_{i,j}$ gives the claim. □

4.3 Regret of one pull

Let $E = E_x \cap E_\theta \cap E_\eta$. Then $\mathbb{P}(E^c) \le 3\delta$.

Lemma 13 (Per-pull regret)**.** *On $E_x \cap E_\eta$, if agent i pulls from bin J at Phase-2 round t, then*

$$r_{i,t} \le \min\left\{2L_f,\ 2w(n_{i,J}(t)) + 2\omega\right\}.$$

This holds for any directions $\hat{\theta}_k$ (all equal or not) and for any set of available bins.

Proof. $r_{i,t} \le 2L_f$ because $|f| \le L_f$ on E_x. For the second bound, assume $n_{i,J} \ge 1$ (otherwise $w = \infty$). Let x^* be the best arm, in bin j^* under $\hat{\theta}_i$. By Lemma 10, $f(\langle x^*, \theta^* \rangle) \le m_{j^*}$.

The best bin looks good. If $n_{i,j^*} = 0$, then $\mathrm{UCB}_{j^*} = \infty$. Since J maximises the UCB, $\mathrm{UCB}_J = \infty$, contradicting $n_{i,J} \ge 1$. So $n_{i,j^*} \ge 1$. Every seen pull of bin j^* has mean $\ge m_{j^*} - \omega$ (Lemma 10), so by Lemma 12,

$$\mathrm{UCB}_{j^*} \ \ge\ (m_{j^*} - \omega) - w(n_{i,j^*}) + w(n_{i,j^*}) \ =\ m_{j^*} - \omega.$$

The chosen bin is not much better than it looks. Every seen pull of bin J has mean $\le m_J$, so

$$\mathrm{UCB}_J \ \le\ m_J + 2w(n_{i,J}).$$

Combine. The pulled arm has mean $\ge m_J - \omega$, and $\mathrm{UCB}_J \ge \mathrm{UCB}_{j^*}$. Therefore

$$r_{i,t} \ \le\ m_{j^*} - (m_J - \omega) \ \le\ (\mathrm{UCB}_{j^*} + \omega) - (\mathrm{UCB}_J - 2w(n_{i,J})) + \omega \ \le\ 2w(n_{i,J}) + 2\omega. \qquad \square$$

4.4 Stale counts

This is the only place where the graph enters the regret bound.

Lemma 14 (Unseen pulls)**.** *Let t be a Phase-2 round, J a bin and $l \ne i$ an agent. The number of pulls by l into J in rounds $e(t) < t' \le t - 1$ is at most $\gamma \max(G_J(e(t)), 1) + 2L - 1$.*

Proof. Let t be in slot $m+1$, so $\beta_m < t \le \beta_{m+1}$ and $e(t) = \max\{e \in \mathcal{E} : e \le \beta_{m-1}\}$. Split the rounds $(e(t), t-1]$ into $(e(t), \beta_{m-1}]$ and $(\beta_{m-1}, t-1]$.

Second part. It has $t - 1 - \beta_{m-1} \le \beta_{m+1} - 1 - \beta_{m-1} = 2L - 1$ rounds, and l pulls once per round.

First part. If $e(t) = \beta_{m-1}$, it is empty. Otherwise $\beta_{m-1} \notin \mathcal{E}$, so no agent raised an alarm at β_{m-1}. In particular, (1) fails for l and J at β_{m-1}:

$$O_{l,J}(\beta_{m-1}) - O_{l,J}(\epsilon_{m-1}) \;\le\; \gamma \max\big(G_J(\epsilon'_{m-1}), 1\big).$$

Here $\epsilon_{m-1} = \max\{e \in \mathcal{E} : e \le \beta_{m-2}\}$. Because $\beta_{m-1} \notin \mathcal{E}$, this equals $\max\{e \in \mathcal{E} : e \le \beta_{m-1}\} = e(t)$. Also $\epsilon'_{m-1} \le e(t)$, and G_J is non-decreasing, so $G_J(\epsilon'_{m-1}) \le G_J(e(t))$. Hence l made at most $\gamma \max(G_J(e(t)), 1)$ pulls into J in $(e(t), \beta_{m-1}]$. □

Lemma 15 (Stale counts)**.** *If the pull of agent i at round t is the k-th Phase-2 pull into bin J (in lexicographic order), then*

$$n_{i,J}(t) \;\ge\; \frac{k - c_{N,\Lambda}}{1 + N\gamma}, \qquad c_{N,\Lambda} := 1 + N(\gamma + 2L) = 1 + N(\gamma + 4\Lambda).$$

Proof. Write $n = n_{i,J}(t)$ and $G = G_J(e(t)) \le n$. The $k-1$ earlier pulls into J are of three kinds:

- seen by agent i: exactly n of them (Lemma 11);
- by another agent $l \ne i$ in rounds $(e(t), t-1]$: at most $(N-1)\big(\gamma \max(G, 1) + 2L - 1\big)$ by Lemma 14;
- by another agent in round t itself: at most $N - 1$.

So $k - 1 \le n + (N-1)\gamma \max(G, 1) + (N-1)\, 2L$.

Case $G \ge 1$. Then $\max(G, 1) = G \le n$, so $k - 1 \le n(1 + N\gamma) + 2NL$, and $n \ge (k - 1 - 2NL)/(1 + N\gamma) \ge (k - c_{N,\Lambda})/(1 + N\gamma)$.

Case $G = 0$. Then $k - 1 \le n + N(\gamma + 2L)$, so $n \ge k - c_{N,\Lambda}$. If $k \le c_{N,\Lambda}$ the claim is trivial. Otherwise $n \ge k - c_{N,\Lambda} \ge (k - c_{N,\Lambda})/(1 + N\gamma)$. □

4.5 The regret theorem

Theorem 16 (Regret of Dec-ZoomSIB-Relay)**.** *Let Assumptions 1 and 2 hold, and let $\delta = 1/(NT)$. Suppose NT_0 satisfies the sample-size condition of Lemma 2. Then for any $\Delta > 0$ and $\gamma > 0$,*

$$\begin{aligned}\mathbb{E}R_T(N) \;\le\; & \underbrace{2L_f N(T_0 + \Lambda)}_{\text{Phase 1}} + \underbrace{4\sqrt{2}\, c_\delta \sqrt{(1 + N\gamma) N_{\mathrm{b}} NT}}_{\text{UCB}} + \underbrace{2L_{f'} \Delta NT}_{\text{discretisation}} \\ & + \underbrace{4L_{f'} X_{\max}\, \varepsilon_0(NT_0)\, NT}_{\text{projection}} + \underbrace{2L_f N_{\mathrm{b}}\, (c_{N,\Lambda} + 1)}_{\text{delay and staleness}} + 6L_f.\end{aligned}$$

Proof. Bad event. $\mathbb{P}(E^c) \le 3\delta = 3/(NT)$, and the regret is always at most $2L_f NT$. This contributes at most $6L_f$.

Phase 1. There are $NT_1 = N(T_0 + \Lambda)$ uniform pulls, each costing at most $2L_f$.

Phase 2 on E. On E_θ, $\varepsilon_{\mathrm{p}} \le X_{\max} \varepsilon_0(NT_0)$. Fix a bin j and let n_j be its final number of Phase-2 pulls. For the k-th of them, Lemmas 13 and 15 give

$$r^{(k)} \;\le\; \min\Big\{2L_f,\; 2c_\delta \sqrt{\tfrac{2(1+N\gamma)}{k - c_{N,\Lambda}}} + 2\omega\Big\},$$

where the second term is used only when $k > c_{N,\Lambda}$. Use $2L_f$ for $k \le c_{N,\Lambda}+1$; at most $c_{N,\Lambda}+1$ such terms. For $k > c_{N,\Lambda}+1$, the numbers $k-c_{N,\Lambda}$ are at least 1 and spaced by 1, so $\sum_k (k-c_{N,\Lambda})^{-1/2} \le \sum_{u=1}^{n_j} u^{-1/2} \le 2\sqrt{n_j}$. Hence

$$\sum_{k=1}^{n_j} r^{(k)} \;\le\; 2L_f(c_{N,\Lambda}+1) + 4\sqrt{2}\, c_\delta \sqrt{(1+N\gamma)\, n_j} + 2\omega\, n_j.$$

Sum over the N_b bins. Cauchy–Schwarz gives $\sum_j \sqrt{n_j} \le \sqrt{N_\mathrm{b} \sum_j n_j} \le \sqrt{N_\mathrm{b} NT}$. Finally, $2\omega NT = 2L_{f'}\Delta NT + 4L_{f'}\varepsilon_\mathrm{p} NT$. □

Corollary 17 (Rate)**.** *Take $\Delta = (NT)^{-1/3}$, $\gamma = \gamma_0/N$ for a constant γ_0, and*

$$NT_0 = \left\lceil \left(L_{f'} X_{\max} \tfrac{4C_\theta d}{\mu^*} \sqrt{\log(2dNT)}\, \tfrac{NT}{L_f} \right)^{2/3} \right\rceil$$

(the choice of the server theorem), at least as large as Lemma 2 requires. Then

$$\mathbb{E}R_T(N) = \tilde{O}\Big((d\,NT)^{2/3} + N\Lambda\,(NT)^{1/3} \Big).$$

If $N\Lambda \le (NT)^{1/3}$, this matches Theorem 5 in N and T up to logarithms.

Proof. $N_\mathrm{b} \le 2X_{\max}(NT)^{1/3} + 1 = \tilde{O}((NT)^{1/3})$, and $c_\delta = \tilde{O}(1)$, so with $N\gamma = \gamma_0$ the UCB term is $\tilde{O}((NT)^{2/3})$. The discretisation term is $2L_{f'}(NT)^{2/3}$. The choice of T_0 equalises $2L_f NT_0$ and the projection term $\propto d\,NT/\sqrt{NT_0}$, and makes both $\tilde{O}((dNT)^{2/3})$. The Phase-1 delay $2L_f N\Lambda$ is dominated by the last term. The last term is $2L_f N_\mathrm{b}(2+\gamma_0+4N\Lambda) = \tilde{O}(N\Lambda(NT)^{1/3})$. □

Remark 18 (What changed from the sketch)**.** The earlier sketch (`server.tex`, Theorem 16) bounded the confidence width of an agent's view, which is a union of N per-agent prefixes, and lost a factor $\sqrt{N}$. Here the view is a common prefix plus one own block (Lemma 11), so Lemma 12 is exactly the server lemma. The price is that agents act on slightly older data: the delay constant is 4Λ instead of Λ. This is the "delayed but consistent statistics" idea used by Dubey & Pentland (2020) and by DLUCB.

Remark 19 (The trigger threshold)**.** The factor $\sqrt{1+N\gamma}$ appears for the same reason as with a server: between two epochs, each of the N agents may add $\gamma\, G_J$ pulls to bin J. E30 measured this cost for the server protocol only. Whether $\gamma = 0.5$ is also nearly free under flooding has not been measured yet.

5 Communication

We count *scalars*: integers or reals sent over one edge in one direction. This is the measure used by Wang et al. (2020).

Lemma 20 (Number of epochs)**.** *For each bin j, let A_j be the set of epochs at which some agent's alarm condition (1) holds for bin j. Then*

$$|A_j| \;\le\; 2\big(2 + \log_{1+\gamma}(NT)\big), \qquad |\mathcal{E}| \;\le\; 1 + 2N_\mathrm{b}\big(2 + \log_{1+\gamma}(NT)\big).$$

Proof. List $A_j = \{a_1 < a_2 < \dots\}$ and let $g_p = G_j(a_p)$. Each a_p is a slot boundary, so $a_{p+2} \geq a_p + 2L$.

Growth over two steps. Let $a_{p+2} = \beta_m$. The alarm at β_m is by some agent l, with $\delta_{l,j}(\beta_m) > \gamma \max(G_j(\epsilon'_m), 1)$. The pulls counted in $\delta_{l,j}(\beta_m)$ happen in rounds $(\epsilon_m, \beta_m]$, after ϵ'_m. So

$$g_{p+2} = G_j(\beta_m) \;\geq\; G_j(\epsilon'_m) + \delta_{l,j}(\beta_m) \;>\; G_j(\epsilon'_m) + \gamma \max\big(G_j(\epsilon'_m), 1\big).$$

Also $a_p \leq \beta_m - 2L = \beta_{m-2}$ and $a_p \in \mathcal{E}$, so $a_p \leq \epsilon'_m$ and $G_j(\epsilon'_m) \geq g_p$. Two cases follow:

- if $g_p \geq 1$, then $g_{p+2} > (1+\gamma)\, G_j(\epsilon'_m) \geq (1+\gamma)\, g_p$;
- if $g_p = 0$, then $g_{p+2} \geq \delta_{l,j} \geq 1$.

Counting. Consider the subsequence $g_1, g_3, g_5, \dots$. After at most one step it is ≥ 1. After that it grows by a factor $> 1+\gamma$ per step, and it never exceeds NT. So it has at most $2 + \log_{1+\gamma}(NT)$ elements. The same holds for $g_2, g_4, \dots$.

Finally, every epoch other than β_0 belongs to some A_j, and only bins that are actually pulled can trigger. □

Theorem 21 (Communication)**.** *The total number of scalars sent by Dec-ZoomSIB-Relay is at most*

$$\underbrace{2|E|\, N\, (d+1)}_{\text{Phase 1}} \;+\; \underbrace{2|E|\, |\mathcal{E}|\, \big(1 + N(3N_{\mathrm{b}} + 1)\big)}_{\text{Phase 2}} \;=\; O\big(|E|\, N\, N_{\mathrm{b}}^2 \log_{1+\gamma}(NT) + |E|\, N\, d\big).$$

In particular it grows only logarithmically in T for fixed N_{b}. By contrast, any protocol in which every agent sends its N_{b}-bin table to its neighbours every round sends $\Theta(|E|\, N_{\mathrm{b}}\, T)$ scalars.

Proof. Under flooding, each message is forwarded by each agent at most once, to each neighbour. So one message crosses each of the $2|E|$ directed arcs at most once.

Phase 1. There are N records, each with $d+1$ scalars.

Phase 2. For each epoch, every agent sends at most one alarm over each arc: either its own alarm or the first one it receives, never both. This is one scalar per arc. It also generates N snapshots. Each snapshot holds at most N_{b} changed bins, at 3 scalars each (index, count increment, sum increment), plus one label. Multiply by $2|E|$ and by the number of epochs from Lemma 20. □

Remark 22. Snapshots can instead be summed up a BFS spanning tree and broadcast back down. Each tree edge then carries at most $3N_{\mathrm{b}} + 1$ scalars per direction per epoch, which removes the factor $N|E|/(N-1)$: $O\big((NN_{\mathrm{b}} + |E|)\, N_{\mathrm{b}} \log_{1+\gamma}(NT)\big)$. We keep flooding in the main statement because it is what the robustness results of Section 6 need. The bound counts every pulled bin as changed in every snapshot, so it is loose. In practice a snapshot contains only the bins the agent pulled since the last epoch.

6 Robustness

The proofs of Sections 4 and 5 use the graph only through Assumption 2. So any communication model with a known delivery bound Λ gets Theorems 16 and 21 with that Λ. It suffices to compute Λ in each case.

Lemma 23 (Delivery time under weaker connectivity)**.** *Assumption 2 holds with the following Λ:*

(i) Directed graphs. *If $\mathcal{G}$ is a static, strongly connected directed graph with directed diameter D, then $\Lambda = D$.*

(ii) Time-varying graphs. *Let the edge set E_s change with the step s. Suppose that for every window of B consecutive steps, the union graph $\bigcup_{s\in window} E_s$ is connected with diameter at most D_B. Use* persistent flooding*: every agent re-sends each message it holds to its current neighbours in every step, until Λ steps after the message was created. Then $\Lambda = B(D_B + 1)$.*

(iii) Link delays. *Let $\mathcal{G}$ have diameter D, and suppose every message sent over a link is delivered within τ steps. Then $\Lambda = \tau D$.*

Proof. (i) This is the proof of Lemma 1, with "neighbour" replaced by "in-neighbour" and distance by directed distance.

(ii) Partial windows waste at most B steps, so consider full windows after the one in which the message was sent. Let U_r be the set of agents that hold the message after r full windows. Take an agent $u \in U_r$ and an edge $\{u, v\}$ that appears at some step of window $r + 1$. Under persistent flooding, u sends the message across that edge at that step, so $v \in U_{r+1}$. Hence U_{r+1} contains U_r and its neighbourhood in the union graph of window $r+1$. That union graph has diameter at most D_B, so $U_{D_B} = V$.

(iii) The induction of Lemma 1, with each hop taking at most τ steps. □

Corollary 24 (Robustness)**.** *Under any model of Lemma 23, with slot length $L = 2\Lambda$ for the corresponding Λ, Theorem 16 and Corollary 17 hold verbatim. In particular, a B-connected time-varying graph costs an additive $\tilde{O}\big(NBD_B(NT)^{1/3}\big)$ in regret. Records arrive late but are never wrong, so the leading $(NT)^{2/3}$ term does not change. Theorem 21 holds verbatim in cases (i) and (iii). In case (ii) it holds with an extra factor Λ, and with $|E|$ replaced by $\max_s |E_s|$, because each message is re-sent for Λ steps.*

Proof. Lemma 7 is the only place where delivery is used, and it uses only Assumption 2. Everything after it depends only on the epoch structure. For communication, persistent flooding sends each message over each present arc at most once per step, for Λ steps. □

Remark 25. Random link failures, such as each link failing independently with probability q per step (one of the settings of E18), give a random delivery time. With persistent flooding, the corollary then applies on the event that every message used is delivered within a chosen Λ. A union bound over messages and over the hops of a shortest path shows that this event has probability at least $1 - 1/(NT)$ when $\Lambda \gtrsim (D + \log(NT))/(1 - q)$. The complement adds a term of order L_f to the bound.

7 Agreement

So far every agent froze the same direction $\hat{\theta}_0$. Approximate Phase-1 methods such as consensus or push-sum leave each agent k with its own direction $\hat{\theta}_k$. Bins are then computed with different projections, but all tables are still added up bin by bin.

Theorem 26 (Cost of disagreement)**.** *Run Phase 2 of Dec-ZoomSIB-Relay with agent k binning by its own $\hat{\theta}_k$, $\|\hat{\theta}_k\|_1 = 1$, on the same grid. Let $\eta = \max_k \|\hat{\theta}_k - \theta^*\|_1$. On the event $\{\eta \le \bar{\eta}\}$, Theorem 16 holds with the projection term replaced by*

$$4L_{f'}X_{\max}\,\bar{\eta}\,NT.$$

In particular, if $\hat{\theta}$ is the exact pooled direction and $\max_k \|\hat{\theta}_k - \hat{\theta}\|_1 \le \eta_{\text{agree}}$, then the extra regret relative to exact pooling is at most $4L_{f'}X_{\max}\,\eta_{\text{agree}}\,NT$. It is zero when Phase 1 pools exactly.

Proof. Lemmas 10, 12, 13 and 15 were stated for arbitrary directions $\hat{\theta}_k$. The directions enter only through $\varepsilon_{\mathrm{p}} = X_{\max} \max_k \|\hat{\theta}_k - \theta^*\|_1$:

- Lemma 10(ii) puts the true index of a bin-j pull by *any* agent in the same widened interval $\tilde{I}_j$. So pooling the bin-j statistics of different agents is still pooling pulls whose means lie in $[m_j - \omega, m_j]$.
- Lemma 12 concerns only noise, which does not depend on the directions.
- Lemma 15 concerns only counts.

The proof of Theorem 16 then goes through with $\varepsilon_{\mathrm{p}} \le X_{\max}\bar{\eta}$. For the second statement, use the triangle inequality $\|\hat{\theta}_k - \theta^*\|_1 \le \|\hat{\theta} - \theta^*\|_1 + \|\hat{\theta}_k - \hat{\theta}\|_1$ and compare with the exact-pooling bound. □

Corollary 27 (An agreement pass is enough)**.** *Suppose Phase 1 uses any method that leaves agent k with a Stein sum b'_k (an estimate of the pooled sum), followed by one flooding pass in which every agent floods its own* local *sum b_k. Then all agents end with the exact pooled direction after Λ more rounds, and Theorem 16 holds with $T_0 + \Lambda$ replaced by $T_0 + 2\Lambda$ in the Phase-1 term.*

Proof. After the flooding pass every agent holds all N local sums (Assumption 2), so all agents compute $\sum_k b_k / \|\sum_k b_k\|_1$. Lemma 9 applies, and so does Theorem 26 with $\eta_{\mathrm{agree}} = 0$. The extra Λ rounds of uniform play cost at most $2L_f N\Lambda$. □

Remark 28 (Link to E28)**.** Theorem 26 is linear in the disagreement and has no threshold. E28 (fig28, panel D) shows regret rising steeply once disagreement exceeds about 10^{-2}. That is consistent, since the bound only becomes comparable to the other terms once $X_{\max}\eta_{\mathrm{agree}}NT$ is of order $(NT)^{2/3}$. E28b confirms Corollary 27: one flooding pass removes the whole penalty of one-step consensus (from $\times 1.25$ to $\times 1.08$ on the ring, and from $\times 1.34$ to $\times 1.09$ on the path).

8 Summary

Result	Statement	Evidence
Thm 5	$\Omega((NT)^{2/3})$ for any graph and protocol	E22
Thm 16, Cor 17	$\tilde{O}((dNT)^{2/3} + N\Lambda(NT)^{1/3})$	E19, E23 (code version)
Thm 21	$O(\lvert E\rvert N N_{\mathrm{b}}^2 \log_{1+\gamma} NT)$ scalars	E15, E20, E23
Cor 24	time-varying, directed, lossy graphs: replace D by Λ	E18
Thm 26	disagreement η costs $\le 4L_{f'} X_{\max}\eta NT$	E28

Open. (a) The additive $N\Lambda N_{\mathrm{b}}$ term: E23 measures a cost 60–150× smaller that grows like $D^{0.65}$. (b) A lower bound that depends on the graph. (c) The trigger cost under flooding (E31). (d) An experiment with the epoch protocol of Section 3, to confirm it behaves like the code version.